Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Koopman Spectral Theory Paper

Benjamin D. Summers <bsummers@adagioinstitute.org> Thu, Sep 3, 2026 at 5:27 PM
To: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>
Cc: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Sara,

Thank you for the review; it earned its keep. Revision 4 is enclosed. I adjudicated each finding individually rather than accepting the set wholesale.

1. Proposition 2.11(c). Accepted — the error is exactly as you diagnosed it. $\frac{1}{2}(J_n - I_n)$ has row sums $(n-1)/2$ and is doubly stochastic only at $n = 3$, so the stated witness failed for every n the clause claimed beyond the one it was proved for. The statement now gives $\frac{1}{2}(J_3 - I_3)$ for $n = 3$ and the padded matrix $\frac{1}{2}(J_3 - I_3) \oplus I_{(n-3)}$ for $n > 3$, with the proof extended: the zero pattern forces $U = U_1 \oplus U_2$, each block is unitary, and $|U_1|^2$ reproduces the $n = 3$ obstruction. A parenthetical records why the unpadded matrix fails, so the trap is marked for the next reader. For your interest: the Coq formalization of this obstruction was always stated for 3×3 — the over-claim lived only in the prose, which is a pattern worth noticing in itself.

2. Remark 2.8, the augmentation passage. Accepted. "No measurable augmentation produces a divisible Markov representation" was false as written — history augmentation Markovizes anything, as you say. The passage now distinguishes unrestricted measurable augmentation (always available) from augmentation respecting the operational constraints of the correspondence setting (what the Bell-type arguments actually rule out), and claims only the latter.

3. §3.2. Accepted; rewritten in two regimes. Time-homogeneous divisible families are elapsed-time semigroups and get Hille–Yosida; genuinely time-dependent divisible families are evolution families and get the Dyson representation. The old text asserted the semigroup structure before making the split, which, as you noted, is wrong in the general case.

4. Generator scoping. Accepted. "No time-ordered-exponential generator of any kind" is now "in the strongly measurable, Bochner-integrable class" in the introduction, with matching language in the abstract and §3.4. Proposition 3.6 proves exactly that and no more.

5. Corollary 4.6, following text. Accepted. D(t,s,r) depends on r, so "invariant under the choice of r" was the wrong sentence; it now says the bound holds for every r, with the strongest form the supremum over $r \in (s,t)$. I also added the interpretation qualifier you suggested, both after the corollary and in the Discussion: the obstruction concerns uniform operator-norm approximation of the two-time family, and says nothing against divisible fits to selected observables or marginals.

6. (R3′) — the one disagreement. Your review reads the hypothesis as absolute continuity of Γ_{ij} and says it should instead be on $\sqrt{\Gamma_{ij}}$. The manuscript's hypothesis has been on the square roots verbatim since the first revision: "for each fixed s the map $t \mapsto \sqrt{(\Gamma_{ij}(t \leftarrow s))}$ is absolutely continuous on [s,T]." Your underlying mathematical point — that absolute continuity of Γ alone would not suffice, because the derivative of √Γ can fail to be integrable

where $\Gamma \to 0$ at the diagonal — is correct, and it is precisely why the hypothesis was placed on the square roots in the first place. Since a careful reader misread it once, I have added a parenthetical to (R3′) stating that the placement is deliberate and why. So: no substantive change, but the text is better for your having tripped on it, and I would rather flag the disagreement openly than fold it silently into the accept pile.

7. Appendix C, Step 1b. Accepted without reservation — the sharpest catch in the review. The frame correspondence is not convex-valued, Michael's theorem does not apply, and the clause invoking it is deleted. Steenrod triviality stands as the primary route and Kato §II.6 as the alternate; the Michael bibitem is removed. This citation misuse survived the examination, my own verification pass, and one prior external audit; it is now recorded in the paper's verification ledger as an audit miss, because pretending otherwise would defeat the ledger's purpose.

8. Appendix C, Step 4. Accepted. Minimality is a per-(t,s) property; the deletion criterion is now the Kraus operators vanishing at that pair, with a note that the (R3′) dichotomy makes the deleted set constant on $\{s < t\}$.

9. Theorem 7.4, hypothesis (C2). Accepted, and your incompatibility argument is now recorded in the paper. Norm continuity up to the diagonal, compactness for $t > s$, and $P(s,s) = I$ jointly force the identity to be compact — unsatisfiable on infinite-dimensional L^2, as you observed — and nothing in the proof ever used the diagonal. (C2) is restated on the open region $\{s < t\}$; Appendix D Step 0 now works on the open simplex; and a new Remark 7.5 gives the argument, names strong continuity as the correct diagonal notion, and notes that the closed reading survives in finite dimensions.

10. Editorial. Accepted as a package. The abstract is cut by roughly forty percent. Proposition 2.6 and Corollary 2.7 are framed explicitly as classical background (tower property; Feller) at both the contributions paragraph and the proposition itself. The introduction now says in one sentence what "Koopman" is doing in the title. The empirical protocol in the Discussion carries the two caveats you asked for — a significant $J(R) > 0$ does not by itself attribute mechanism, and the estimated subspaces are stable only up to Davis–Kahan/Wedin factors, both now cited. The grid dilation is identified as the standard repeated-interaction (collision-model) structure, citing the Ciccarello–Lorenzo–Giovannetti–Palma review. One editorial item I considered and left alone: the remark preceding Theorem 4.5 already presents part (a) as concentration extracted by standard Schmidt machinery rather than as a novelty claim, so I kept that text — if you think it still oversells, tell me where.

The bibliography nets out at −1 (Michael) +3 (Davis–Kahan; Wedin; Ciccarello et al.), and the machine-checked layer is untouched by the whole round: every one of your confirmed findings lived in statements, hypotheses, or citations rather than in a computation, which is consistent with where the previous rounds' findings lived too.

On Convex Perturbation: I'd welcome the same treatment. Fair warning that it cites this paper's Theorem 4.5, Proposition 4.12, and Theorems 5.4 and 7.4, so revision 4 is the right companion text; I'll send the current post-examination revision of the perturbation paper if you don't already have it. Same terms as this round: every finding gets an individual verdict and a reasoned reply, including the ones I reject.

Thank you again.

Ben
